

Exercise 2: E-commerce Platform Search Function

CODE:

```
class Product {
    int productId;
    String productName;
    String category;
    Product(int productId, String productName, String category) {
        this.productId = productId;
        this.productName = productName;
        this.category = category;
    }
    void display() {
        System.out.println("Product ID : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Category : " + category);
    }
}

public class Main
// Linear Search
public static Product linearSearch(Product[] products, String name) {
    for (Product p : products) {
        if (p.productName.equalsIgnoreCase(name)) {
            return p;
        }
    }
    return null;
}

// Binary Search
public static Product binarySearch(Product[] products, String name) {
```

```

int left = 0;
int right = products.length - 1
while (left <= right) {
    int mid = (left + right) / 2;
    int compare = name.compareToIgnoreCase(products[mid].productName);
    if (compare == 0) {
        return products[mid];
    } else if (compare > 0) {
        left = mid + 1;
    } else {
        right = mid - 1;
    }
}
return null;
}

public static void main(String[] args) {
    Product[] products = {
        new Product(105, "Keyboard", "Accessories"),
        new Product(101, "Laptop", "Electronics"),
        new Product(104, "Monitor", "Electronics"),
        new Product(102, "Mouse", "Accessories"),
        new Product(103, "Phone", "Electronics")
    };
    System.out.println("Linear Search:");
    Product result1 = linearSearch(products, "Phone");
    if (result1 != null) {
        result1.display();
    } else {
        System.out.println("Product not found.");
    }
}

```

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    }
    System.out.println();
    // Sorted array for Binary Search
    Product[] sortedProducts = {
        new Product(105, "Keyboard", "Accessories"),
        new Product(101, "Laptop", "Electronics"),
        new Product(104, "Monitor", "Electronics"),
        new Product(102, "Mouse", "Accessories"),
        new Product(103, "Phone", "Electronics")
    };
    System.out.println("Binary Search:");
    Product result2 = binarySearch(sortedProducts, "Phone");
    if (result2 != null) {
        result2.display();
    } else {
        System.out.println("Product not found.");
    }
}
}

```

OUTPUT:

```

Linear Search:
Product ID : 103
Product Name : Phone
Category : Electronics

Binary Search:
Product ID : 103
Product Name : Phone
Category : Electronics

```